Higher National Diploma in Engineering (Civil/Mechanical/BSE)
1st year, Second Semester Examination – 2022
CE 1207/ BSE 1206 – Strength of Materials 1

Instructions for candidates:	No. of questions : 05
Answer four questions only.	No. of pages : 03
All questions carry **equal** marks.	Time : 02 hours

01)

a) Define the term hardness of a material. (02 marks)

b) Name three different hardness testing methods. (03 marks)

c) Draw stress vs strain curve for a ductile material under tensile test. Mark all critical points. (03 marks)

c) Define the following terms with relevant equations.

i) Linear strain (01 marks)

ii) Lateral strain (01 marks)

iii) Poisson's ratio (01 marks)

d) A steel bar 3 m long, 50 mm wide and 20 mm thick is subjected to an axial full of 150 kN in the direction of its length. Find the changes in length, width and thickness of the bar. Take E = 180 GPa and Poisson's ratio = 0.3. (14 marks)

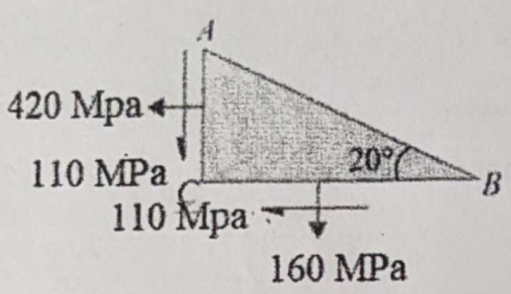
02)

a) Explain the terms, principal planes and principal stresses. (05 marks)

b) A plane element in a boiler is subjected to tensile stresses of 420 MPa on one plane and 160 MPa on the other at the right angle to the former as shown in the figure. Each of the above stresses is accompanied by a shear stress of 110 MPa such that when associated with the major tensile stress tends to rotate the element in the anticlockwise direction. Find,

a) Principal stress and their direction (10 marks)

b) Maximum shearing stress and direction of the plane on which they act. (10 ma



03)

a) Define following terms

i) Shear force (02 ma

ii) Bending moment (02 m

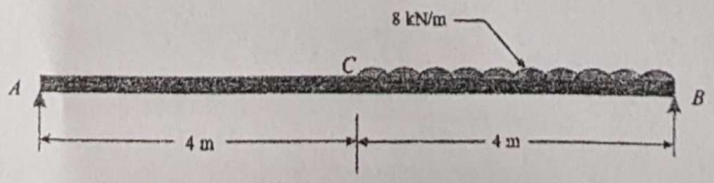
b) A simply supported beam 8m long is carrying a uniformly distributed load of 8 kN/m over a length of 4 m from the right end.

i) Find the values of support reactions at A & B. (06 m

ii) Derive expressions for the shear force and the bending moment for each segment of the beam (06 m

iii) Draw the shear force and bending moment diagrams. (06 m

iv) Find the maximum bending moment and its location (03 m



04) a) Derive expressions for maximum slope and maximum deflection of a simply supported beam. (10 marks)

b) A simply supported solid square (150 mm x 150 mm) beam of 4 m in length is subjected to a central load of 15 kN. If the modulus of elasticity of the beam is 180 GPa, Find

i) Maximum deflection of the beam (08 marks)

ii) Maximum slope of the beam (07 marks)

05)

a) Derive an expression for maximum torque (T) that can be transmitted via a solid shaft with diameter (D). Take the modulus of rigidity of the shat material (C) and maximum allowable shear stress (τ). (08 marks)

b) What are the assumptions you made to derive the above expression? (03 marks)

d) Find the maximum torque that can be safely applied to a solid shaft of 90 mm diameter. The permissible angle of twist is 4 degrees in a length of 10 m and shear stress should not exceed 50 MPa. Take modulus of rigidity is 90 GPa. (14 marks)